Course Syllabus

DATA 422: Advanced Data Science Methods and Ethics — Fall 2026

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Cal Poly Humboldt — Fall 2026

Course information

Course Number: DATA 422
Course Title: Advanced Data Science Methods and Ethics
Units: 4
Semester: Fall 2026
Mode of Instruction: In person
Prerequisite: DATA 322
Minimum grade for major: C-

Lecture

Section: DATA 422-10 (Class Number 42778)
Days/Time: Monday, Wednesday, Friday, 11:00–11:50 a.m.
Location: NR 224

Laboratory (Activity)

Section: DATA 422-11 (Class Number 42779)
Day/Time: Tuesday, 11:00 a.m.–12:50 p.m.
Location: BSS 313

Project 3 presentations (final-exam slot)

Friday, December 18, 2026, 10:20 a.m.–12:10 p.m., NR 224.

Instructor

Instructor: Dr. Rosanna Overholser
Email: rho3@humboldt.edu
Office: BSS 334
Phone: 707-826-4020

Office hours: Wednesdays 12:30–2:30 p.m. in the Learning Center (first floor of the library); Thursdays 11:00–11:50 a.m. in BSS 334; or by appointment.

Please use your Cal Poly Humboldt email and check Canvas regularly for course communications.

Catalog description

This course provides a treatment of advanced topics in Data Science. Topics include: (i) Privacy and Ethics, (ii) Multivariate Statistics, (iii) Big Data and Cloud Computing, and (iv) Deep Learning. The course will draw from numerous case studies and applications, with a practical rather than theoretical emphasis.

Required materials

- **Textbook:** *The Hundred-Page Language Model Book* (free online), plus readings provided in Canvas.
- **Textbook**: *How to Argue and Win Every Time: At Home, at Work, in Court, Everywhere* by Gerry Spence (used copies can be purchased very cheaply)
- **Technology:** A computer for work outside lab (personal, Data Science program, or library checkout). If you need a laptop, consider Cal Poly Humboldt's Reusable Office Supply Exchange (R.O.S.E.) program.

Course load

DATA 422 has 3 units of lecture and 1 unit of activity. Expect about **5 hours in class** and at least **7 hours out of class** each week.

Assessment and grading

Component	Weight
Quizzes	40%
Project 1	20%
Project 2	20%
Project 3	20%

How assessments work

- **Project 3** is written in the university **final-exam time slot** (Friday, December 18, 10:20 a.m.–12:10 p.m., NR 224).
- **Labs** are ungraded Colab notebooks (objectives, syntax, demos, self-check exercises). They are **not** a grade.

Stepping stones are project checkpoints; they count toward the related project, not lab.

Weekly quizzes are **paper, in class**, in the **first 10 minutes of Friday lecture** (15 points). They draw on that week's **lab**, **lecture**, and assigned **reading or videos**. Study banks are on the site; Canvas is the reminder and the gradebook.

Letter grades (nearest percent): 93–100 (A); 90–92 (A-); 87–89 (B+); 83–86 (B); 80–82 (B-); 77–79 (C+); 73–76 (C); 70–72 (C-); 60–69 (D); below 60 (F).

See Assessments for links as materials are posted.

Important dates

Date	Event
August 24	Classes begin
September 7	Labor Day (no class)
November 11	Veterans Day (no class)
November 23–27	Fall break (no class)
December 11	Last class meeting
December 18	Project 3 presentations, 10:20 a.m.–12:10 p.m., NR 224

See the schedule for the week-by-week calendar.

Course structure

Meetings include discussions and small-group activities based on readings distributed beforehand. Projects practice applying course content. Typical project deliverables include shared code of ethics with reflection, a presentation (Project 3 / final slot), and a written report with code appendix—exact prompts will be posted under projects.

Missed or late work

Quizzes can be made up in Rosanna's Office Hours. Project stepping stones or deliverables can be turned in late by making an arrangement with Rosanna. The only exceptions are the deliverables for projects 2 and 3, which will involve class participation for which there is no substitute.

Academic integrity

Students are responsible for knowing and abiding by Cal Poly Humboldt's academic honesty policy. Students caught cheating will receive zero points and be reported to Student Judicial Affairs. Working in small groups on projects and when preparing for quizzes is encouraged, but work submitted for grading must be your own.

AI policy

You may use AI (for example ChatGPT or Claude) to help you learn the materials, perform data analyses, or work on projects. For graded work completed outside class:

1. If you used AI, acknowledge it briefly in the document (name the service; detailed prompts are not required).
2. You are responsible for the correctness and appropriateness of everything you turn in (AI-assisted or not).

The instructor may use AI to improve course materials (for example checking whether an assignment is college-level or clarifying quiz wording). AI will **not** be used to grade your work, and AI detection tools will not be used to check your work.

Campus policies and resources

Syllabus addendum: campus resources and policies

Learning outcomes

Program / student learning outcomes (summary)

Students will demonstrate computational skills to extract and visualize data; statistical knowledge to build models and ensure valid analysis; contemporary data-oriented skills with ethical considerations; and effective communication for diverse audiences.

Course learning outcomes

Upon successfully completing this course, you will be able to apply advanced data science methods and ethical reasoning in practical case studies spanning privacy, multivariate statistics, big data / cloud computing, and deep learning.